

Placement Empowerment Program
Cloud Computing and DevOps Centre

Use Cloud Storage

Create a storage bucket on your cloud platform and upload/download files. Configure access permissions for the bucket.

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1. Introduction and Overview

Cloud storage is a scalable and flexible solution for storing and managing data on cloud platforms. This PoC explores the process of creating a cloud storage bucket, uploading and downloading files, and configuring access permissions using AWS S3, Google Cloud Storage (GCS), and Azure Blob Storage. The implementation of cloud storage ensures accessibility, security, and cost-effectiveness for managing digital assets.

2. Objective

- To create a storage bucket on a cloud platform that can store various types of files.
- To upload and download files efficiently to and from the storage bucket.
- To configure appropriate access permissions to maintain security while allowing authorized users access.
- To evaluate the usability and performance of different cloud storage solutions.

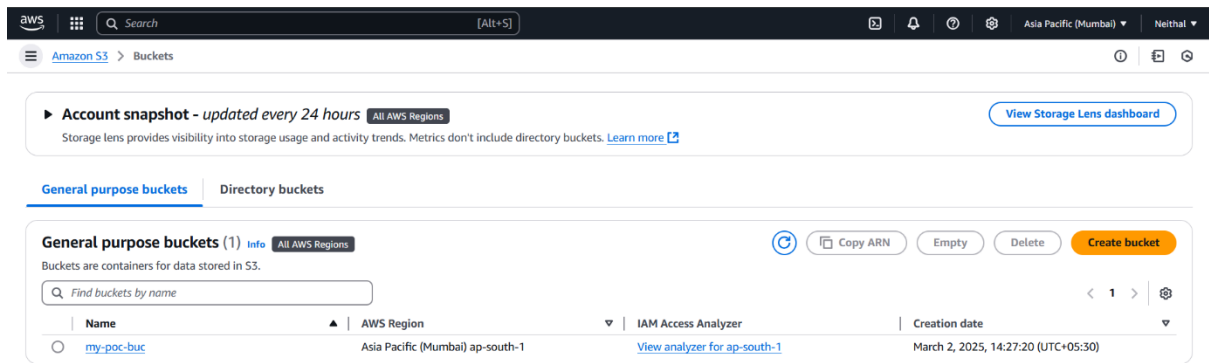
3. Importance

Cloud storage provides numerous benefits, including scalability, reliability, and accessibility from any location. It allows businesses and individuals to store and retrieve data without the need for on-premise hardware. By implementing appropriate security policies, unauthorized access can be prevented while ensuring that permitted users can efficiently manage stored data. Additionally, cloud storage enhances disaster recovery strategies, as data is not dependent on a single local system but is stored redundantly across multiple locations.

Steps for Execution

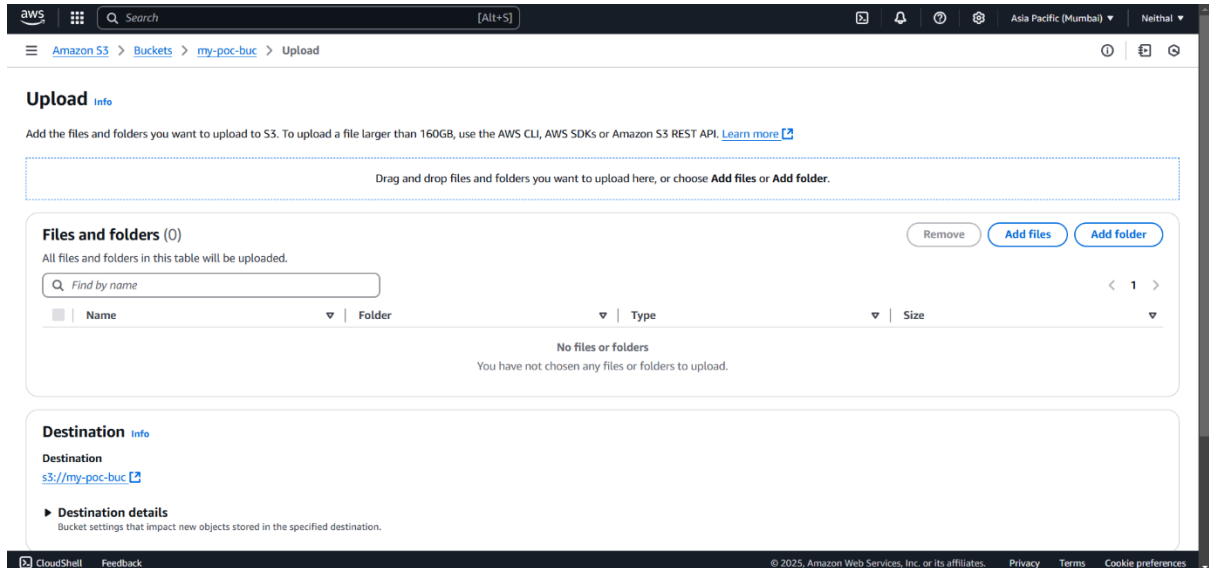
Step 1: Create a Cloud Storage Bucket

The first step involves logging into the selected cloud provider's platform and navigating to the storage management section. A new storage bucket is created by providing a unique name and selecting the appropriate geographic region to optimize access speeds. Other settings, such as storage class, data redundancy, and versioning, can be configured based on specific use cases before finalizing the creation of the bucket.



Step 2: Upload a File to the Bucket

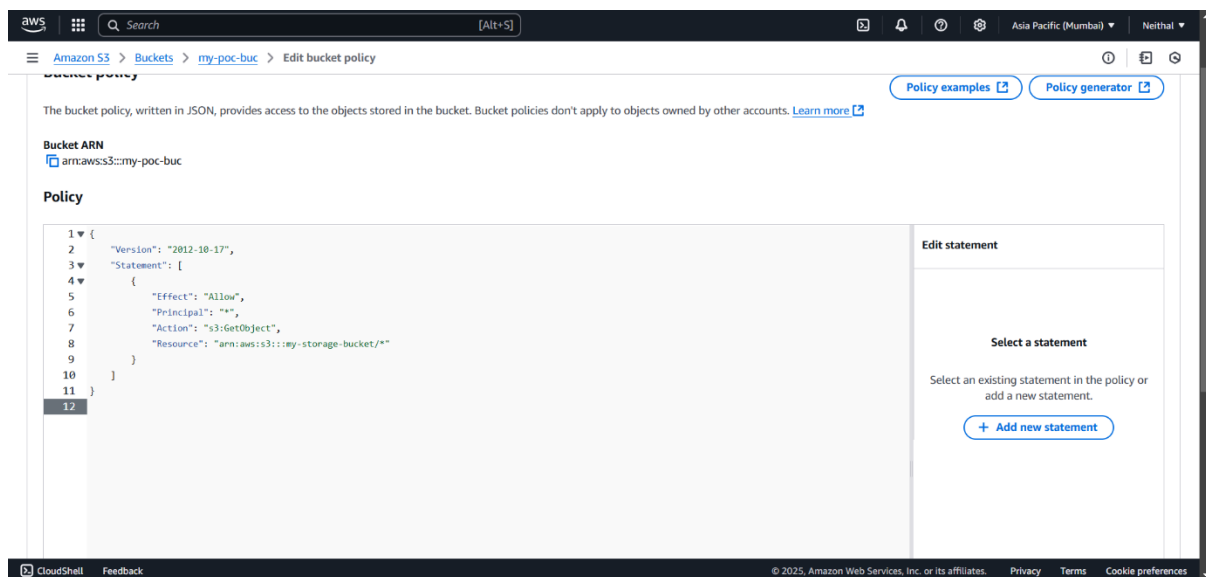
Once the storage bucket is set up, users can upload files by selecting them from their local system. The uploaded files are stored within the bucket and can be categorized and managed through the cloud platform's interface. The system may provide options to manage file metadata, apply storage lifecycle rules, and enable encryption for added security.



Step 3: Configure Access Permissions

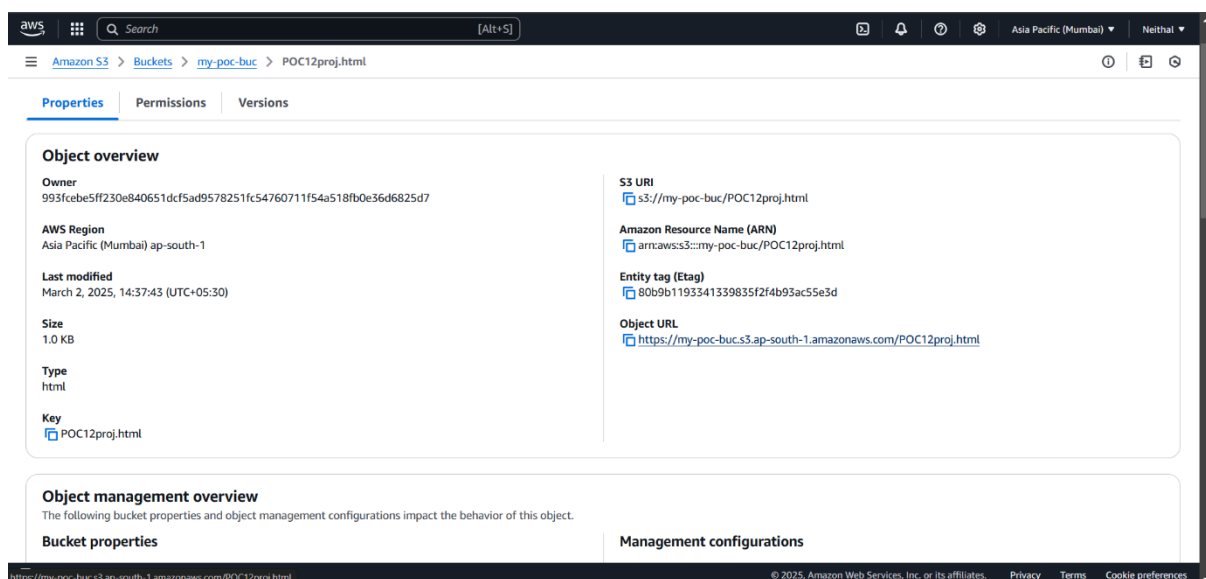
Configuring access permissions ensures that only authorized users can interact with the stored data. This involves modifying settings within the cloud platform's permission management system. Permissions can be set at different levels, such as granting full administrative access, read-only access, or restricting access to specific users or applications. Depending on security

requirements, public access can be enabled or restricted to prevent unauthorized data exposure. Role-based access control (RBAC) policies may also be implemented for better security management.



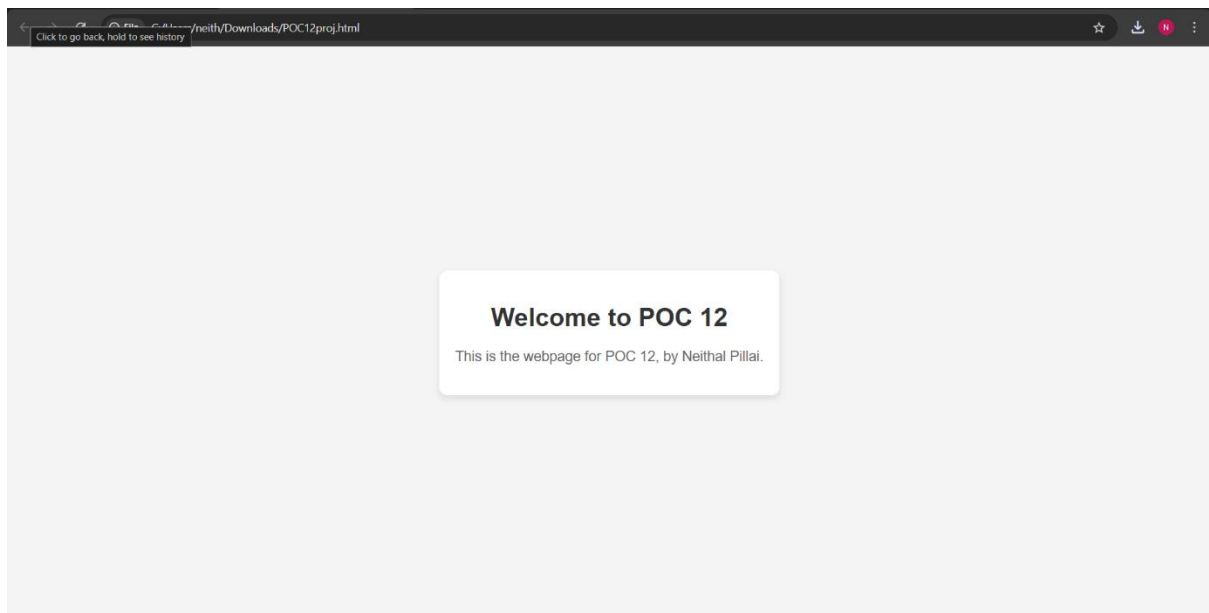
Step 4: Download a File from the Bucket

After uploading a file, users may need to retrieve it from the cloud storage. The download process can be performed through the cloud provider's web interface, where files can be selected and saved to a local machine. In addition to manual downloads, cloud providers offer automation tools that allow scheduled or programmatic retrieval of stored data. This ensures that data can be efficiently accessed as required.



Step 5: Validate and Test the Implementation

Once the setup is complete, thorough testing is performed to verify that all storage operations function as expected. This includes ensuring that files can be uploaded and downloaded without corruption, confirming that access permissions are correctly enforced, and reviewing security settings to prevent unauthorized modifications. Additionally, performance metrics such as upload/download speeds and cost estimates are analyzed to evaluate the efficiency of the storage solution.



5. Conclusion

This PoC successfully demonstrates the creation and management of cloud storage buckets across different cloud platforms. By following a structured approach, it is possible to establish a secure and efficient cloud storage solution that meets data storage needs. The implementation also highlights the importance of proper access control and data management strategies to ensure security and usability.

What I Learned

- Cloud storage platforms offer flexible and scalable data management solutions.
- Proper access control is essential to prevent unauthorized data access while enabling collaboration.
- Cloud-based storage reduces the need for physical infrastructure and enhances data recovery options.

- Implementing storage automation improves efficiency and operational workflows.